# Chapter 4

# Electrons in a periodic potential: Bloch's theorem

We know that the lattice sites form a repetitive array, so the potential felt by an electron moving in it will be periodic as well, this means (as we have discussed during X-ray scattering) we can write potential as a Fourier sum over all reciprocal lattice vectors

## 4.1 Derivation of the theorem

$$\begin{aligned} H &= T + V \\ &= T + \sum_{\mathbf{G}} v_{\mathbf{G}} e^{i\mathbf{G}.\mathbf{r}} \end{aligned} \tag{4.1}$$

Let's assume we know all the components $v_{\mathbf{G}}$ and want to solve for the the eigenvalues and eigenfunctions. We know that if the potential was zero, then the solutions would have been free particle like, now as a consequence of the existence of the potential, many plane waves states $|\mathbf{k}\rangle$ must be mixed, so

$$|\psi\rangle = \sum_{\mathbf{k}} c_{\mathbf{k}} |\mathbf{k}\rangle \tag{4.2}$$

where

$$\langle \mathbf{r}|\mathbf{k}\rangle = \frac{1}{\sqrt{L^3}} e^{i\mathbf{k}.\mathbf{r}} \tag{4.3}$$

To solve for $c_{\mathbf{k}}$ is the obvious target. We let $H$ act on the *ket* $|\psi\rangle$ , which we assume to be an eigenstate of H, and then left multiply with the state *bra* $\langle \mathbf{k}'|$.

$$\begin{aligned} \langle \mathbf{k}'|H|\psi\rangle &= \sum_{\mathbf{k}} c_{\mathbf{k}} \langle \mathbf{k}'|T|\mathbf{k}\rangle + \sum_{\mathbf{k}} c_{\mathbf{k}} \langle \mathbf{k}'|V|\mathbf{k}\rangle \\ \therefore \quad E c_{\mathbf{k}'} &= \frac{\hbar^2}{2m} k'^2 c_{\mathbf{k}'} + \sum_{\mathbf{k}} \sum_{\mathbf{G}} c_{\mathbf{k}} v_{\mathbf{G}} \delta_{\mathbf{k}',\mathbf{k}+\mathbf{G}} \\ \therefore \quad 0 &= \left(\frac{\hbar^2}{2m} k'^2 - E\right) c_{\mathbf{k}'} + \sum_{\mathbf{G}} c_{\mathbf{k}'-\mathbf{G}} v_{\mathbf{G}} \end{aligned} \tag{4.4}$$

The sum is over the reciprocal lattice vectors $\mathbf{G}$. This has the form of an eigenvalue equation. But we have only one equation for a large number of unknowns -all the $c_{\mathbf{k}-\mathbf{G}}$. In principle there can be an infinite number of them.

Now notice a few important points

1. Eqn 4.4 connects a state $\mathbf{k}$ with states that can be reached by reciprocal lattice translations. *i.e.* The potential only mixes the states $|\mathbf{k}\rangle$, $|\mathbf{k} - \mathbf{G_1}\rangle$, $|\mathbf{k} - \mathbf{G_2}\rangle$ and so on.

2. But there are some states which cannot be taken to one another by reciprocal lattice translations: take any two states in the first Brillouin zone, that are not on the zone boundary.

3. The previous two points together imply that $\mathbf{k}$ from the first Brillouin zone can be used to label a wavefunction that will contain $|\mathbf{k}\rangle$, $|\mathbf{k}-\mathbf{G_1}\rangle$, $|\mathbf{k}-\mathbf{G_2}\rangle$.... These wavefunctions are the Bloch wavefunctions.

Now we complete the solution. In the process of getting to eqn. 4.4 we could have left multiplied with another state $\langle\mathbf{k}'-\mathbf{K}|$ where $\mathbf{K}$ is any reciprocal lattice vector. Then we would get the equation

$$\left(\frac{\hbar^2}{2m}(\mathbf{k}'-\mathbf{K})^2 - E\right) c_{\mathbf{k}'-\mathbf{K}} + \sum_{\mathbf{G}} c_{\mathbf{k}'-\mathbf{K}-\mathbf{G}} v_{\mathbf{G}} = 0 \tag{4.5}$$

Hence

$$\left(\frac{\hbar^2}{2m}(\mathbf{k}'-\mathbf{K})^2 - E\right) c_{\mathbf{k}'-\mathbf{K}} + \sum_{\mathbf{G}'} c_{\mathbf{k}'-\mathbf{G}'} v_{\mathbf{G}'-\mathbf{K}} \;=\; 0$$

where we have written $\mathbf{G}' = \mathbf{K}+\mathbf{G}$, remembering that $\mathbf{k}'$ and $\mathbf{K}$ are fixed vectors for a particular row. The primes are now unnecessary in the summation indices.

$$\left(\frac{\hbar^2}{2m}(\mathbf{k}-\mathbf{K})^2 - E\right) c_{\mathbf{k}-\mathbf{K}} + \sum_{\mathbf{G}} c_{\mathbf{k}-\mathbf{G}} v_{\mathbf{G}-\mathbf{K}} = 0$$

$\mathbf{G}$ and $\mathbf{K}$ run over the same set of (reciprocal lattice) vectors. Now we see the following:

1. There will be as many solutions as the number of reciprocal lattice vectors we keep.

2. Each $\mathbf{k}$ vector (not a reciprocal lattice vector) in the first Brillouin zone gives a matrix for us to solve. The $E(\mathbf{k})$ relation thus has as many branches as the dimension of the matrix.

We can see that the wavefunction must have the form

$$\begin{aligned}
|\psi_{\mathbf{k}}\rangle &= \sum_{\mathbf{G}} c_{\mathbf{G}} |\mathbf{k}-\mathbf{G}\rangle \\
\langle\mathbf{r}|\psi_{\mathbf{k}}\rangle \equiv \psi_{\mathbf{k}}(\mathbf{r}) &= \sum_{\mathbf{G}} c_{\mathbf{G}} e^{i(\mathbf{k}-\mathbf{G}).\mathbf{r}} \\
&= e^{i\mathbf{k}.\mathbf{r}} \sum_{\mathbf{G}} c_{\mathbf{G}} e^{-i\mathbf{G}.\mathbf{r}} \\
&= e^{i\mathbf{k}.\mathbf{r}} u(\mathbf{r}) && (4.6) \\
\therefore \quad \psi_{\mathbf{k}}(\mathbf{r}+\mathbf{R}) &= e^{i\mathbf{k}.(\mathbf{r}+\mathbf{R})} u(\mathbf{r}+\mathbf{R}) \\
&= e^{i\mathbf{k}.(\mathbf{r}+\mathbf{R})} \sum_{\mathbf{G}} c_{\mathbf{G}} e^{-i\mathbf{G}.\mathbf{r}} e^{-i\mathbf{G}.\mathbf{R}} && (4.7) \\
&= e^{i\mathbf{k}.\mathbf{R}} \psi(\mathbf{r}) \qquad \left(\because e^{-i\mathbf{G}.\mathbf{R}} = 1\right)
\end{aligned}$$

Since where $\mathbf{k}$ is in the first Brillouin zone. This is Bloch's theorem in many equivalent forms. The function $u(\mathbf{r})$ has the symmetry of the direct lattice.

### 4.1.1 Translation invariance and Bloch's theorem

The Hamiltonian of the lattice is invariant under a translation by any lattice vector $\mathbf{R}$. We construct the translation operator in terms of the momentum operator $\mathbf{p}$ as

$$\begin{aligned}
T(\mathbf{R}) &= e^{i\mathbf{R}.\mathbf{p}/\hbar} && (4.8) \\
T(\mathbf{R})^{-1} H T(\mathbf{R}) &= H \\
\therefore [T(\mathbf{R}), H] &= 0 && (4.9)
\end{aligned}$$

Since $\mathbf{p}$ is hermitian, $T(\mathbf{R})$ is unitary. The commutator tells us that $T(\mathbf{R})$ and $H$ has simultaneous eigenfunctions. So if $|\Psi\rangle$ is an eigenfunction of $H$, it should also satisfy the two following conditions

$$T(\mathbf{R})\psi(\mathbf{r}) = c(\mathbf{R})\psi(\mathbf{r}) \quad (4.10)$$
$$T(\mathbf{R})\psi(\mathbf{r}) = \psi(\mathbf{r}+\mathbf{R}) \quad (4.11)$$

Now what is $c(\mathbf{R})$ ?

$$\begin{aligned}\langle\mathbf{k}|T(\mathbf{R})|\psi\rangle &= \langle\psi|T^\dagger(\mathbf{R})|\mathbf{k}\rangle^* \\ &= \left(\int d^3\mathbf{r}\ \psi^*(\mathbf{r}).e^{-i\mathbf{R}.\mathbf{p}/\hbar}.e^{i\mathbf{k}.\mathbf{r}}\right)^* \\ &= \left(\int d^3\mathbf{r}\ \psi^*(\mathbf{r}).e^{i\mathbf{k}.(\mathbf{r}-\mathbf{R})}\right)^* \\ &= e^{i\mathbf{k}.\mathbf{R}}\langle\mathbf{k}|\psi\rangle \end{aligned} \quad (4.12)$$

Then left multiply 4.10 with $\langle\mathbf{k}|$ and put together the result with 4.12. This leads to

$$\left(c(\mathbf{R}) - e^{i\mathbf{k}.\mathbf{R}}\right)\langle\mathbf{k}|\psi\rangle = 0 \quad (4.13)$$

Using this and eqn 4.11 you should be able to reproduce the properties of the Bloch functions.

### 4.1.2 Significance of k

The vector $\mathbf{k}$ looks like momentum ($\hbar\mathbf{k}$), but it is not. To see this, let us calculate the momentum of a particle in a Bloch state. We need the expectation value of the momentum operator $\mathbf{p} = -i\hbar\nabla$

$$\langle\mathbf{p}\rangle = \langle\psi_\mathbf{k}| - i\hbar\nabla|\psi_\mathbf{k}\rangle = \hbar\mathbf{k} + \hbar\sum_\mathbf{G}\mathbf{G}|c_{\mathbf{k}+\mathbf{G}}|^2 \quad (4.14)$$

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PROBLEM : Prove eqn. 4.14.

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To distinguish $\hbar\mathbf{k}$ from momentum, we will call it the *crystal momentum*. The significance of this will be clear when we write the equations of semiclassical dynamics of the electron.

#### Response of a Bloch electron to an Electric field

Consider a chain of atoms in the form of a loop (1D) so that periodic boundary conditions are easy to apply. However it is precisely because of the periodic boundary condition an extra electric field is difficult to add to the potential $V(x)$. The extra field (resulting from connecting a voltage source to measure electric current for example) is hard to accommodate if we write it as $V = -eEx$. We use another way of introducing an extra field. Recall

$$\mathbf{E} = -\frac{\partial\mathbf{A}}{\partial t} - \nabla V \quad (4.15)$$

We introduce the extra $E$ using a time dependent $A = -Et$. So the Hamiltonian including $A(t)$ becomes:

$$H = \frac{(p+eA)^2}{2m} + V(x) \quad (4.16)$$

This has *time dependent* eigenfunctions (of the Bloch form) and eigenenergies, which we call $|\Psi(x,t)\rangle$ and $\varepsilon(t)$ to distinguish it from the electric field E.

$$\left(\frac{(p+eA)^2}{2m} + V(x)\right)|\Psi(x,t)\rangle = \varepsilon(t)|\Psi(x,t)\rangle \quad (4.17)$$

We now introduce an unknown parameter $\Lambda$ and write

$$\begin{aligned} |\Psi(x,t)\rangle &= e^{i\Lambda x}|\Phi(x,t)\rangle && (4.18)\\ \therefore (p+eA)|\,\Psi(x,t)\rangle &= \hbar\Lambda e^{i\Lambda x}|\Phi(x,t)\rangle - i\hbar e^{i\Lambda x}\frac{\partial}{\partial x}|\Phi(x,t)\rangle + eAe^{i\Lambda x}|\Phi(x,t)\rangle && (4.19)\end{aligned}$$

The first and last terms will cancel if we choose

$$\hbar\Lambda + eA = 0 \tag{4.20}$$

This choice results in the simplification

$$(p+eA)|\,\Psi(x,t)\rangle = -i\hbar e^{i\Lambda x}\frac{\partial}{\partial x}|\Phi(x,t)\rangle = e^{i\Lambda x}p|\Phi(x,t)\rangle \tag{4.21}$$

Eqn. 4.17 can now be written as (a few intermediate steps are left for you to work out...)

$$\left(\frac{p^2}{2m} + V\right)|\Phi\rangle = \varepsilon(t)|\Phi\rangle \tag{4.22}$$

$|\Phi\rangle$ must have solutions of the Bloch form such that

$$\Phi(x) = e^{ikx}u_k(x) \tag{4.23}$$

Using this and the periodic boundary conditions we get:

$$\begin{aligned} \Psi(x+L) &= \Psi(x)\\ \therefore e^{-ieA(x+L)/\hbar}e^{ik(x+L)}u_k(x+L) &= e^{-ieA(x)/\hbar}e^{ik(x)}u_k(x)\\ \therefore e^{-ieAL/\hbar}e^{ikL} &= 1\\ \therefore eEt/\hbar + k &= 2n\pi/L\\ \therefore \hbar\frac{dk}{dt} &= -eE && (4.24)\end{aligned}$$

The label $k$ *behaves like* momentum. The result is a striking consequence of periodicity and elementary quantum mechanics put together! Later on we will treat this problem with an electric and magnetic field as well.

### 4.1.3 Origin of the band gap: solving the matrix equation

Let us for the moment consider a simplified "toy" case to appreciate an (perhaps the most) important consequence of a periodic potential.

1. We consider a 1-dimensional case (chain with a period $a$)
2. We consider a periodic potential

$$V(x) = 2V_0\cos\frac{2\pi x}{a} = V_0e^{i\frac{2\pi}{a}x} + V_0e^{-i\frac{2\pi}{a}x} \tag{4.25}$$

   which means that only two (reciprocal lattice) components of the potential contribute:

$$V_{\frac{2\pi}{a}} = V_{-\frac{2\pi}{a}} = V_0 \tag{4.26}$$

So as to write out the matrix of eqn 4.5 we need to order all the reciprocal vectors according to some rule (which we can frame for ourselves). In the 1D case we can write all the RLVs as:

$$V_n = n.\frac{2\pi}{a} \qquad (\text{n} = ......-2,-1,0,1,2,....) \tag{4.27}$$

So the matrix resulting from the eqn. 4.6 is a tridiagonal matrix, a *part* of which looks like (assuming $\hbar^2/2m = 1$) for simplicity

$$\begin{vmatrix} \left(k+3\frac{2\pi}{a}\right)^2-E & V_{-\frac{2\pi}{a}} & 0 & 0 & 0 & 0 & 0 \\ V_{\frac{2\pi}{a}} & \left(k+2\frac{2\pi}{a}\right)^2-E & V_{-\frac{2\pi}{a}} & 0 & 0 & 0 & 0 \\ 0 & V_{\frac{2\pi}{a}} & \left(k+\frac{2\pi}{a}\right)^2-E & V_{-\frac{2\pi}{a}} & 0 & 0 & 0 \\ 0 & 0 & V_{\frac{2\pi}{a}} & k^2-E & V_{-\frac{2\pi}{a}} & 0 & 0 \\ 0 & 0 & 0 & V_{\frac{2\pi}{a}} & \left(k-\frac{2\pi}{a}\right)^2-E & V_{-\frac{2\pi}{a}} & 0 \\ 0 & 0 & 0 & 0 & V_{\frac{2\pi}{a}} & \left(k-2\frac{2\pi}{a}\right)^2-E & V_{-\frac{2\pi}{a}} \\ 0 & 0 & 0 & 0 & 0 & V_{\frac{2\pi}{a}} & \left(k-3\frac{2\pi}{a}\right)^2-E \end{vmatrix} = 0 \tag{4.28}$$

In the notation of eqn. 4.6, **G** varies along a row, **K** is kept fixed. For every **K** we have a new row. If we had more components of the potential ($e.g. V_{\frac{4\pi}{a}}, V_{-\frac{4\pi}{a}}$) then another band would appear symmetrically about the diagonal line. The matrix is necessary symmetrical and "band diagonal".

### Zero potential case : the "empty lattice"

Now let's consider the apparently trivial case (but there is purpose!) where all the potential components are zero. Then the eignevalues must be the free electron values

$$E = \frac{\hbar^2}{2m}\left(k - V_{n\frac{2\pi}{a}}\right)^2 \qquad \text{with} \quad -\frac{\pi}{a} < k < \frac{\pi}{a} \tag{4.29}$$

It might be a little counter-intuitive to see how the plot with shifted parabolas look. We are used to seeing one continuous parabola ($E \propto k^2$) as the dispersion, because in the free particle case there is no lattice. But if we now shift any vector from outside the first Brillouin zone into the first zone by reciprocal vector translations then this is how it will look. This way of plotting it is called the "reduced zone" scheme and is the most commonly used way of plotting bandstructure. If the lattice is 2d or 3d (Kittel gives an example) then these plots can look quite striking, but it is perfectly logical way of plotting it. We will see soon that the effect of a small non-zero potential would be to smooth the many sharp corners and crossings in the plot as seen in Fig. 4.1.

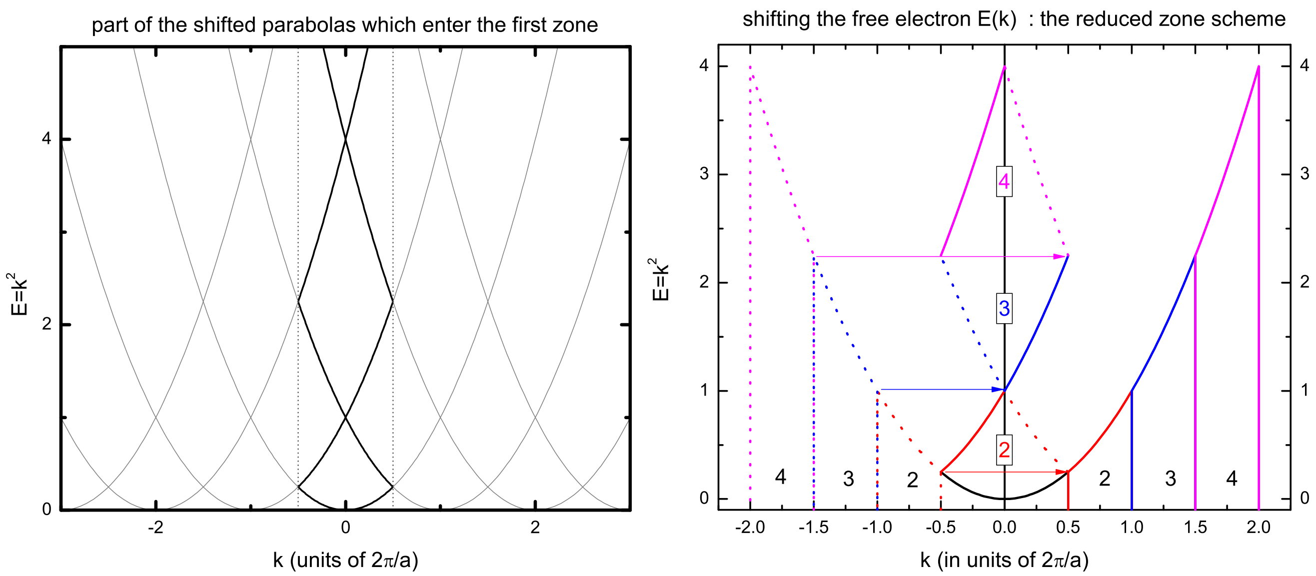


Figure 4.1: (left)Free electron $E(k)$ in the reduced zone scheme. The equations $E = \left(k - n.\frac{2\pi}{a}\right)^2$, and the part which enters the first Brillouin zone. The part in the first zone is in black, the outside parts are grayed out. (right) Another similar plot , showing how the same can be obtained from $E = k^2$ by shifting different parts by different reciprocal lattice translations.

**Turning on a small potential**

We can see that at the point where the shifted parabolas cross, two eignevalues are degenerate. The first case happens at $k = \frac{\pi}{a}$. Here the $E = k^2$ and $E = \left(k - \frac{2\pi}{a}\right)^2$ branches give the same value if $V = 0$. What happens if we turn on a small $V$? We can treat the problem with perturbation (and we will later) but first we just solve for the eigenvalues of the matrix equation 4.28. The plot shows a crucial prediction of the Bloch equation. If we have a potential with a Fourier component ($v_{\mathbf{G}}$) then a gap opens at $\mathbf{G}/2$ These are precisely the zone boundaries. Note that the sharp crossings of Fig. 4.1 have been "rounded off". It turns out that this is a generic feature of turning on a small interaction potential and you will see this in many situations.

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PROBLEM : A quick (though bit handwaving) way of seeing what happens when a small potential is "turned on" is to solve the 2×2part of the matrix eqn 4.28:

$$\begin{vmatrix} k^2 - E & V_0 \\ V_0 & \left(k - \frac{2\pi}{a}\right)^2 - E \end{vmatrix} = 0 \tag{4.30}$$

where $V_{-\frac{2\pi}{a}} = V_{\frac{2\pi}{a}} = V_0$. Now solve for the eignevalues, this should give the behaviour near $k = \pi/a$. You should be able to show that the two eigenvalues at $k = \pi/a$ differ by $2V_0$.

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**Extended and repeated zone schemes**

Exactly the same data can be plotted in some different ways. The repeated zone scheme simply means that we emphasize the periodicity of the solution that $E(\mathbf{k}) = E(\mathbf{k} + \mathbf{G})$. The extended zone scheme shows very clearly the deviation from the free electron parabola. The data used to plot Fig. 4.2 has been plotted in these two schemes in the next figure, Fig .4.3.

### 4.1.4 Band structure as a perturbation problem

The potential is being turned on slowly and we traced the evolution of the energy eignevalues starting with the free electron case. Surely, we could have treated it as a perturbation problem. The answer is "yes", but there are some degeneracies to be aware of. This fact and the connection with Bloch's theory will now become clear.

Time-independent perturbation tells us that the unperturbed state kets $|\mathbf{k}\rangle$ will now be mixed. The point to note is that only those states of the form $|\mathbf{k} + \mathbf{G}\rangle$ will mix with $|\mathbf{k}\rangle$. We write the perturbed state kets to *first* order and perturbed energies to *second* order.

$$\begin{aligned} |\psi_{\mathbf{k}}\rangle &= |\mathbf{k}\rangle + \sum_{\mathbf{G}\neq\mathbf{0}} \frac{\langle\mathbf{k}|V|\mathbf{k}+\mathbf{G}\rangle}{E^0(\mathbf{k}) - E^0(\mathbf{k}+\mathbf{G})} |\mathbf{k}+\mathbf{G}\rangle \\ &= \sum_{\mathbf{G}\neq\mathbf{0}} \frac{v_{\mathbf{G}}}{E^0(\mathbf{k}) - E^0(\mathbf{k}+\mathbf{G})} |\mathbf{k}+\mathbf{G}\rangle \end{aligned} \tag{4.31}$$

$$\begin{aligned} E(\mathbf{k}) &= E^0(\mathbf{k}) + \cancel{\langle\mathbf{k}|V|\mathbf{k}\rangle} + \sum_{\mathbf{G}\neq\mathbf{0}} \frac{|\langle\mathbf{k}|V|\mathbf{k}+\mathbf{G}\rangle|^2}{E^0(\mathbf{k}) - E^0(\mathbf{k}+\mathbf{G})} \\ &= E^0(\mathbf{k}) + \sum_{\mathbf{G}\neq\mathbf{0}} \frac{|v_{\mathbf{G}}|^2}{E^0(\mathbf{k}) - E^0(\mathbf{k}+\mathbf{G})} \end{aligned} \tag{4.32}$$

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PROBLEM : We have dropped the first order energy shift in eqn. 4.32. Why are we justified in doing this? When will this be incorrect?

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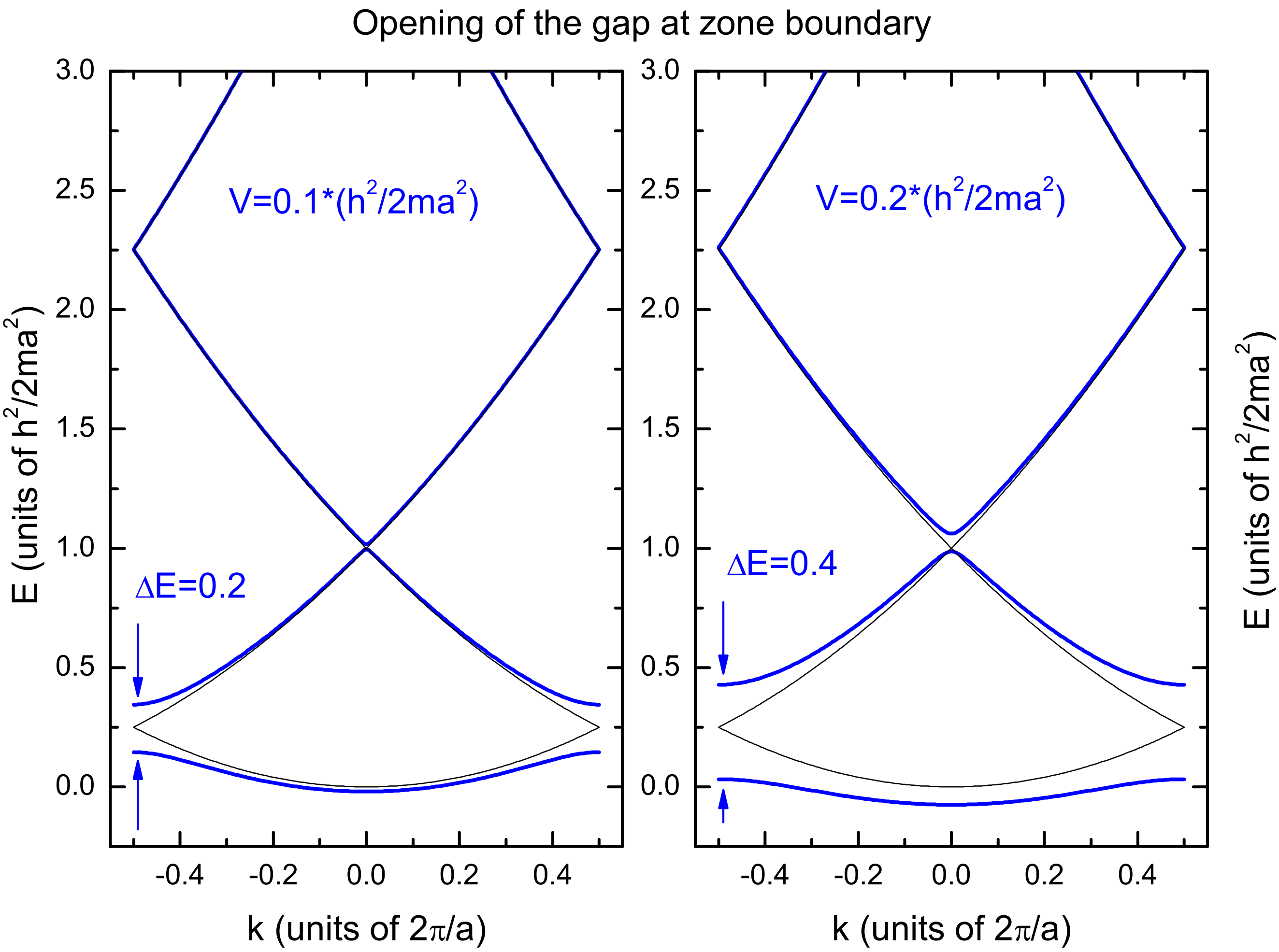


Figure 4.2: $E(k)$ in the reduced zone scheme, with a single non-zero component of the potential as in the equation 4.28. The energies have been scaled with $E = h^2/2ma^2$, which is 4 times the kinetic energy at the zone boundary, to make the plot. Similarly $k$ has been scaled by $2\pi/a$. The black lines show the free electron result with no potential turned on. Note that the potential also affects higher bands but much less..

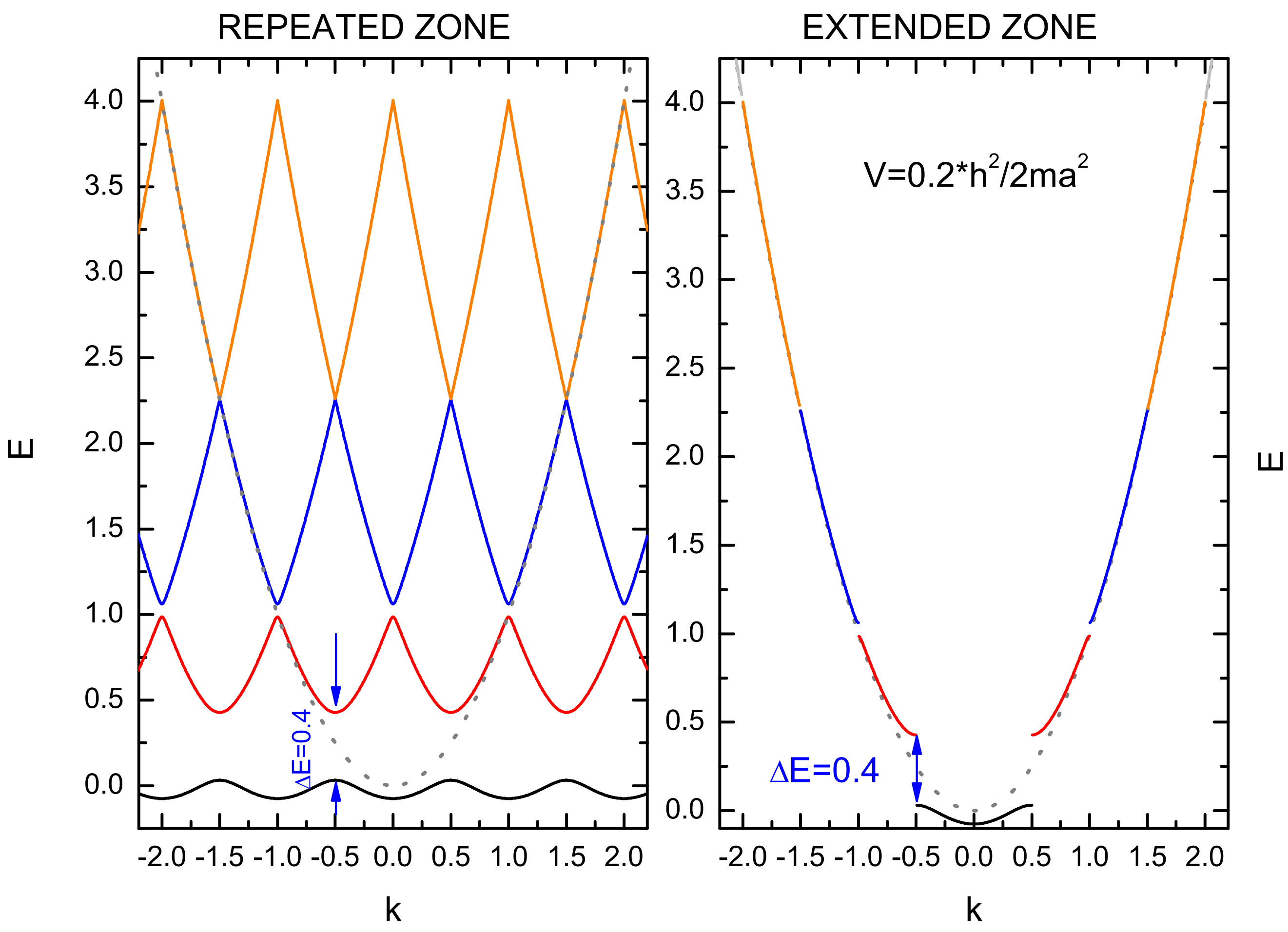


Figure 4.3: $E(k)$ in the repeated and extended zone schemes. The data used is exactly the same as in Fig. 4.2.

1. We do not need to make the sum in eqn. 4.31 run over all $\mathbf{k}' \neq \mathbf{k}$ but only some of them because $\langle \mathbf{k}|V|\mathbf{k}'\rangle = 0$ unless $\mathbf{k}' = \mathbf{k} + \mathbf{G}$. The consequence of all this is that the first order result gives us something of the Bloch form.

2. But if $E^0(\mathbf{k}) - E^0(\mathbf{k}+\mathbf{G}) = 0$ then the denominator will vanish and this will not work. This happens when two states are degenerate. This is precisely where two parabolas in Fig. 4.1 intersect. These are the zone boundaries.

3. At the zone boundaries, the energy shift is no longer of second order, this is then a degenerate first order problem. The solution (see any quantum mechanics text) for the energy shifts are to be obtained by taking the eigenvalues of the matrix $V_{ij}$ between the degenerate states. This is precisely what we did when we solved for (in a previous problem):

$$\begin{vmatrix} E^0(\mathbf{k}) - E & \langle \mathbf{k}|V|\mathbf{k}+\mathbf{G}\rangle \\ \langle \mathbf{k}+\mathbf{G}|V|\mathbf{k}\rangle & E^0(\mathbf{k}+\mathbf{G}) - E \end{vmatrix} = 0 \tag{4.33}$$

### 4.1.5 The Kronig-Penny model

A simple model of a periodic potential is shown in the figure. The well-barrier-well sequence is more realistic than a single fourier component. It is a very useful "toy model" for understanding how the energy bands and band gaps in a solid arises.

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PROBLEM : Revise the single electron in a finite quantum well. Consider a potential given by

$$\begin{aligned} V &= 0 \quad \text{for} \quad -w < x < 0 \\ &= V_0 \qquad \text{otherwise} \end{aligned}$$

Write the wavefunctions for bound states, piecewise as follows

$$\begin{aligned} \Psi_1 &= Ae^{i\alpha x} + Be^{-i\alpha x} \quad \text{for} \quad -w < x < 0 \\ \Psi_2 &= Ce^{-\beta x} \qquad \text{for} \quad x > 0 \\ \Psi_3 &= De^{\beta x} \qquad \text{for} \quad x < -w \end{aligned} \tag{4.34}$$

where $\alpha$ and $\beta$ are given by:

$$\begin{aligned} \alpha^2 &= \frac{2mE}{\hbar^2} \\ \beta^2 &= \frac{2m(V_0 - E)}{\hbar^2} \end{aligned} \tag{4.35}$$

The wavefn and its derivative must be continuous at the two boundaries of the potential well. Show that the solutions are obtained by solving the set of linear equations:

$$\begin{pmatrix} 1 & 1 & -1 & 0 \\ i\alpha & -i\alpha & \beta & 0 \\ e^{-i\alpha w} & e^{i\alpha w} & 0 & -e^{-\beta w} \\ i\alpha e^{-i\alpha w} & -i\alpha e^{i\alpha w} & 0 & -\beta e^{-\beta w} \end{pmatrix} \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix} = 0 \tag{4.36}$$

There are 5 unknowns, but one of the coefficients can be chosen arbitrarily and normalisation will take care of it. Setting the determinant to zero you can show that the wavevector $\alpha$ can be obtained from one of the two conditions

$$\begin{aligned} \tan\frac{\alpha w}{2} &= \frac{\beta}{\alpha} \\ \tan\frac{\alpha w}{2} &= -\frac{\alpha}{\beta} \end{aligned} \tag{4.37}$$